A MORPHISM OF SCHEMES IS FLAT AND LOCAL COMPLETE INTERSECTION EXACTLY WHEN IT IS SYNTOMIC

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18244-18255 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18257-18303, the complete author proof (45 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side only (B-side proof incomplete per ruling). First pillar of the L16201 descent proof.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma syntomic-lci; Algebra, lemma relative-global-complete-intersection-conormal; More on Algebra, lemmas regular-koszul-regular and homotopy-koszul; Divisors, lemma relative-regular-immersion-flat-in-neighbourhood; chapter lemma lci-properties.

Source reference. Original lemma 00SV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18150–18157. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0663: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 062F: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07D3: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18303–18303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A morphism is flat and lci if and only if it is syntomic. Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

(1) *f is flat and a local complete intersection morphism, and*
(2) *f is syntomic.*

Proof. Working affine locally this is More on Algebra, Lemma 07D3. We also give a more geometric proof.

Assume (2). By Morphisms, Lemma 01UE for every point x of X there exist affine open neighbourhoods U of x and V of $f(x)$ such that $f|_U : U \to V$ is standard syntomic. This means that $U = \mathrm{Spec}(R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)) \to V = \mathrm{Spec}(R)$ where $R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ is a relative global complete intersection over R. By Algebra, Lemma 00SV the sequence $f_1, \ldots, f_c$ is a regular sequence in each local ring $R[x_1, \ldots, x_n]_{\mathfrak{q}}$ for every prime $\mathfrak{q} \supset (f_1, \ldots, f_c)$. Consider the Koszul complex $K_\bullet = K_\bullet(R[x_1, \ldots, x_n], f_1, \ldots, f_c)$ with homology groups $H_i = H_i(K_\bullet)$. By More on Algebra, Lemma 062F we see that $(H_i)_{\mathfrak{q}} = 0$, $i > 0$ for every $\mathfrak{q}$ as above. On the other hand, by More on Algebra, Lemma 0663 we see that H_i is annihilated by $(f_1, \ldots, f_c)$. Hence we see that $H_i = 0$, $i > 0$ and $f_1, \ldots, f_c$ is a Koszul-regular sequence. This proves that $U \to V$ factors as a Koszul-regular immersion $U \to \mathbf{A}^n_V$ followed by a smooth morphism as desired.

Assume (1). Then f is a flat and locally of finite presentation (Lemma 069H). Hence, according to Morphisms, Lemma 01UE it suffices to show that the local rings $\mathcal{O}_{X_s,x}$ are local complete intersection rings. Choose, locally on X, a factorization $f = \pi \circ i$ for some Koszul-regular immersion $i : X \to P$ and smooth morphism $\pi : P \to S$. Note that $X \to P$ is a relative quasi-regular immersion over S, see Divisors, Definition 063S. Hence according to Divisors, Lemma 063U we see that $X \to P$ is a regular immersion and the same remains true after any base change. Thus each fibre is a regular immersion, whence all the local rings of all the fibres of X are local complete intersections. □